

# KOA MARIE GERVAIS NELLY

Lead AI Engineer & Data Architect | Founder @ Archi Cam AI

## EXECUTIVE SUMMARY

Lead AI Engineer & Data Architect (Google Developer Program Member), leveraging Google Antigravity to design neuro-symbolic autonomous multi-agent systems, Neo4j GraphRAG, and sovereign production MLOps. Specialized in deterministic architectures and verifiable AI. Founder of Archi Cam AI (official applicant for Google Africa Applied AI Lab), bridging civil engineering methodology, math rigor, and product vision.

## FLAGSHIP AI PROJECTS

### Archi Cam AI — *Agentic AI SaaS & 5D BIM*

- Official Applicant Google Africa Applied AI Lab. Built with Google Antigravity (Gemma 4 12B, Gemini, BAEL 91).
- Automated Excel BOQs in <45s (−99.2% time, MLflow  $R^2=0.9872$ ) and 3D renders via Imagen 3 + ControlNet.

### K1-MATHINFO (v3.0.0) — *Sovereign Multi-Agent AI & OKF Certification*

- Sovereign DMI system (Univ. of Ngaoundéré): 470 theses, 19 M1 projects, 1,366 Neo4j nodes.
- 6-agent LangGraph, hybrid retrieval RRF k=60 + Cross-Encoder, SHA-256 No-LLM cert, 77/77 tests (100%).

### Sovereign.BI Agentic — *Enterprise Security & Agentic BI*

- NL-to-SQL/Graph engine (PostgreSQL pgvector, Neo4j N10S, <5s latency) with ABAC guardrails + SHAP Sentinel.

### Dataset Automator & VigieSahel — *MLOps Pipeline & Climate AI Impact*

- Dataset Automator: Google Antigravity (Google Cloud Hackathon) — TabFM, BigQuery DataFrames, EU AI Act.
- VigieSahel: −35% crop loss, +14d epidemic forecast (XGBoost  $R^2>94\%$ , Supabase, MLflow).

## PROFESSIONAL EXPERIENCE

### AI Lead & Data Science Consultant — *2025 – Present*

#### Independent Projects & Enterprises | Douala, CM

- Sovereign AI agents, Neo4j GraphRAG pipelines, PostgreSQL, EDA and executive decision dashboards.

### Aviation Security Officer (AVSEC) — *2018 – Present*

#### CCAA (Cameroon Civil Aviation Authority)

- Threat assessment, secure access control, regulatory compliance audits (ICAO Annex 17).

## EDUCATION & CERTIFICATIONS

### M.Sc. in Applied Artificial Intelligence — *2025 – 2027 [In Progress]*

#### University of Ngaoundéré

- ML & Bayesian Statistics, Data Engineering & Neo4j, Computer Vision & Robotics, Cybersecurity & Blockchain, Production MLOps.

### B.Sc. in Civil Engineering (Building Option) — *2015 – 2016*

#### ISTDI / IUC Douala

- Structural calculations (BAEL 91), quantity surveying, and construction project management.

## HONORS & APPLIED AI RECOGNITIONS

- Google Africa Applied AI Lab (Accra, 2026):** Official candidacy — Archi Cam AI platform.
- Google Cloud #AllThingsAgentic Hackathon:** Dataset Automator v4.0 (Google Antigravity, TabFM, WIT, bigframes, MCT).



## CONTACT & PROFILES

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🏆 Google Developer Program

## AI & LLM STACK

Google Antigravity IDE ■■■■■  
Google Gemma 4 (12B) ■■■■■  
Gemini 2.5 / 1.5 Pro ■■■■■  
Google TabFM (Tabular) ■■■■■  
LangGraph Multi-Agents ■■■■■  
Firebase Genkit ■■■■■  
Neo4j GraphRAG — Agent K1

## DATA & GRAPHS

Neo4j / Cypher Graph ■■■■■  
PostgreSQL / pgvector ■■■■■  
Google BigQuery ■■■■■ □  
Supabase Realtime ■■■■■ □  
Pandas / NumPy ETL ■■■■■

## DEV & MLOPS

Python 3.11+ / MLOps ■■■■■  
FastAPI / Next.js 14 ■■■■■ □  
MLflow & Data Drift ■■■■■ □  
SHAP Sentinel Audit ■■■■■  
Docker & Streamlit ■■■■■  
IfcOpenShell (5D BIM) ■■■■■ □

## SECURITY & AUDIT

- Quorum 4-Eyes Governance
- OKF v0.2 SHA-256 No-LLM
- EU AI Act (RSASSA-PSS)
- AVSEC (ICAO Annex 17)

## LANGUAGES

French — Native / Fluent  
English — Technical / Pro

## KEY ASSETS

- Dual Competence: AI & Civil Eng.
- Aviation Security & Crisis (CCAA)
- Mathematical Rigor & Guardrails